Nexus Platform - Incident Response & Troubleshooting Manual

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SECTION 1: INCIDENT SEVERITY CLASSIFICATION

- SEV-1 (Critical Outage): Core service down, data corruption risk, or outage affecting >5% of active users. Target response time: <10 minutes.

- SEV-2 (Major Degradation): Degraded search performance, ingestion delays >15 minutes, or isolated regional latency. Target response time: <30 minutes.

- SEV-3 (Minor Defect): Non-critical bugs with available workarounds. Target response time: <4 hours.

SECTION 2: HIGH DATABASE CONNECTION SATURATION RUNBOOK

Symptoms: API endpoints return HTTP 500 or ‘asyncpg.exceptions.TooManyConnectionsError‘.

Diagnostic Steps:

1. Check active pgBouncer connection count: ‘SHOW POOLS;‘

2. Inspect long-running blocking queries in PostgreSQL:

‘SELECT pid, now() - query_start AS duration, query FROM pg_stat_activity WHERE state != ’idle’ ORDER BY duration DESC;‘

3. Terminate stuck rogue query: ‘SELECT pg_terminate_backend(pid);‘

4. If saturation persists, scale pgBouncer max client connections from 100 to 250 in Helm values.

SECTION 3: QDRANT VECTOR SEARCH LATENCY SPIKE RUNBOOK

Symptoms: Hybrid search endpoint latency exceeds 1.5 seconds.

Diagnostic Steps:

1. Inspect Qdrant CPU and memory consumption via Grafana dashboard.

2. Verify if HNSW background indexing is saturating CPU threads.

3. Check memory swap utilization. If memory exceeds 85%, adjust ‘hnsw_config.ef_construct‘ or scale cluster node size.

4. Execute collection optimization trigger: ‘POST /collections/hybrid_knowledge_base/update_params‘ to compact segments.